

4.1 Forces: Turning Effects & Equilibrium

Question Paper

Course	CIEA Level Physics
Section	4. Forces, Density & Pressure
Topic	4.1 Forces: Turning Effects & Equilibrium
Difficulty	Hard

Time allowed: 50

Score: /39

Percentage: /100

Question 1a

A disc made from thin card has radius 42 mm and is pinned to a board so that it hangs vertically in equilibrium.

Three forces act in the plane of the disc. Two forces act tangentially on the circumference with magnitude F_1 . Another acts at a point half way between the pin and edge, at an angle of 30° from the radial line with magnitude F_2 .

The arrangement is shown in Fig. 1.1.

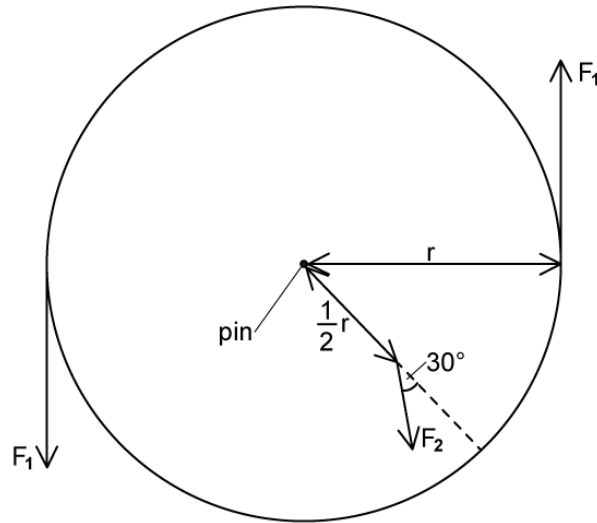


Fig. 1.1

The disc has negligible weight and all frictional forces can be ignored.

For the arrangement shown, express the force F_2 in terms of the forces F_1

[4 marks]

Question 1b

Hence state the value of F_2 if the couple produced by the pair of forces with magnitude F_1 is 0.26 Nm.

[3 marks]**Question 1c**

Determine the magnitude of the force of the pin on the disc.

[1 mark]

Question 2a

In a double-decker bus there are two passenger decks, stacked one above the other.

The dimensions of an older, 'Routemaster' bus are:

Height = 4.38 m

Width = 2.44 m

Mass = 7.47 tons

To ensure safety, buses are tested for stability using a tilting platform as shown in Fig. 1.1.

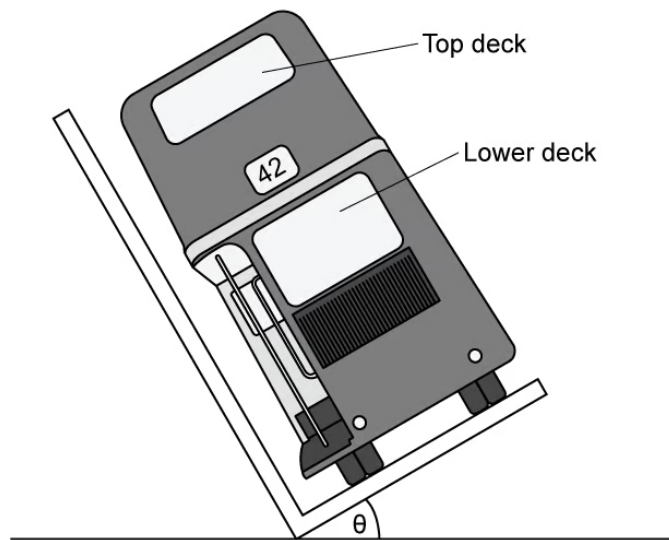


Fig. 1.1

Weights are attached to the underside of the bus, then the angle the bus makes with the horizontal is gradually increased until the bus begins to topple to the left.

(i)

State the reason that the bus would topple to the left.

[1]

(ii)

Explain the effect of adding weights to the underside of the bus.

[3]

[4 marks]

Question 2b

The bus is first tested completely empty, and is found to tilt to an angle of 40° from the horizontal before it starts to tip.

Determine the height above the ground of the centre of mass of the bus when it is empty and normally parked.

[2 marks]**Question 2c**

To be considered safe to use on the road the bus must be safe when carrying passengers.

To test this the top deck is weighted with iron blocks or sandbags to the equivalent of 60 passengers, while the lower deck is left empty.

The average passenger is assumed to have a mass of 65 kg and their centres of mass can be considered to act at a height of 3.28 m above the road.

$$1 \text{ ton} = 1000 \text{ kg}$$

For this safety test

- (i)
Calculate the new height of the centre of mass of the bus above the road.
- (ii)
Suggest why the test is done with the top deck weighted to capacity and the bottom deck left empty.

[2]**[1]****[3 marks]**

Question 2d

To pass the safety test the bus must be able to tilt to 28° when loaded as in part (c).

(i)

Deduce whether the bus has passed the safety test.

[2]

(ii)

The safety testers imagine that the bus had 60 passengers upstairs, no passengers downstairs and that **all** the passengers on the top deck stood up simultaneously. This would shift their centres of mass by a mean distance of 40 cm higher.

Explain whether this will make the bus fail its safety test.

[1]

[3 marks]

Question 3a

In the 1st century CE Greek-Egyptian engineer Hero of Alexandria described a simple steam engine, which was at least 100 years old when he wrote about it.

Now commonly called a Hero's engine, the machine is a steam turbine which can be made to rotate by boiling water in a sealed vessel beneath it. Two pipes channel the steam produced through nozzles which project outwards.

Fig. 1.1. shows Hero's engine in operation.

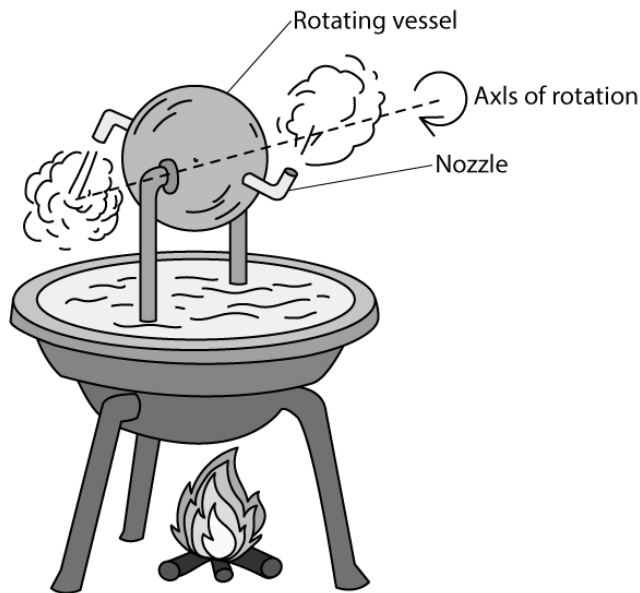


Fig. 1.1.

Explain why the vessel rotates around its axis.

i)
Refer to energy stores and transfers.

[3]

ii)
Refer to forces, naming any relevant laws.

[5]

[8 marks]

Question 3b

A version of Hero's engine can be made for demonstration purposes by puncturing the sides of an empty soda can, adding water then suspending it over a Bunsen burner, as shown in Fig. 1.2.

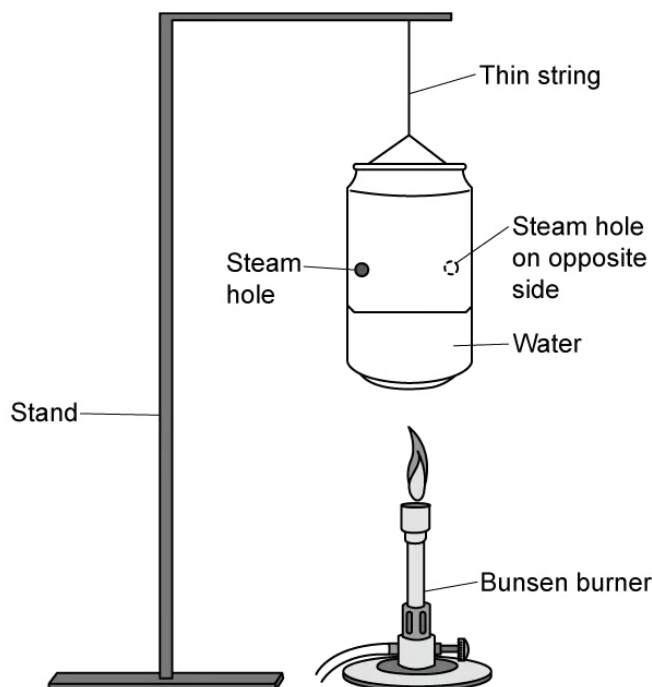


Fig. 1.2.

The holes are made by sticking a pin directly into the metal, then pulling the pin sideways to create an angled vent. This causes the jet of steam to leave the can tangentially, as shown in Fig. 1.3.

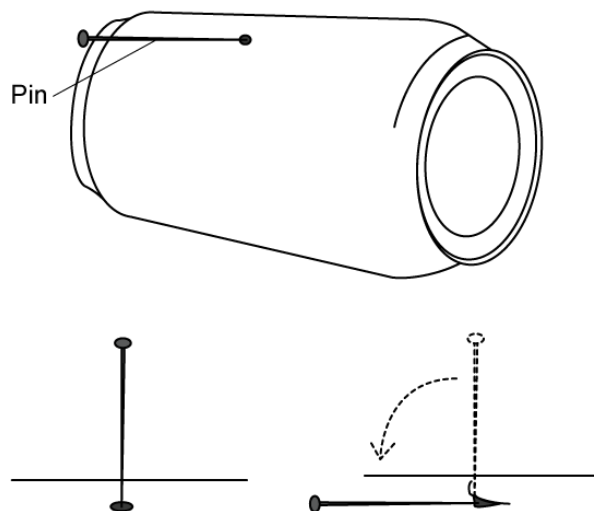


Fig. 1.3.

Explain why simply puncturing holes straight into the can will not produce the rotation seen in the original Hero's engine.

[4 marks]**Question 3c**

A student tries to make a Soda Can Hero's Engine with two pairs of steam holes rather than one.

The dimensions of the can are

height = 115 mm

diameter = 66 mm

Each of the four jets exerts a force of 0.60 N and the holes create jets of steam which are at 45° to the surface of the can.

Calculate the torque experienced by the can.

[4 marks]**Question 3d**

Without adding any further holes, discuss an adjustment that could be made to the can to make it spin more quickly.

[3 marks]

Model Answers: Hard

1a

(a) For the arrangement shown, express the force F_2 in terms of the forces F_1 :

Write expressions for the torque of the couple and the torque of the single force:

- Torque of the couple, $\tau_1 = 2 F_1 r$ [1 mark]
- Torque of the single force, $\tau_2 = (F_2 \times \sin 30) \times \frac{r}{2} = 0.25 F_2 r$ [1 mark]

The disc is in equilibrium, so apply the principle of moments:

- $\tau_1 = \tau_2$ [1 mark]
- $2 F_1 r = 0.25 F_2 r$
- $F_2 = 8 F_1$ [1 mark]

[Total: 4 marks]

1b

(b) Hence state the value of F_2 if the couple produced by the pair of forces with magnitude F_1 is 0.26 Nm:

Calculate the value of F_1 :

- $\tau_1 = F_1 \times r$
- $F_1 = \frac{\tau_1}{r} = \frac{0.26}{(42 \times 10^{-3})}$ [1 mark]
- $F_1 = 6.19$ **or** 6.2 N [1 mark]

Calculate the value of F_2 :

- $F_2 = 8 F_1 = 50 \text{ N (2 s.f.)}$ [1 mark]

[Total: 3 marks]

1c

(c) Determine the magnitude of the force of the pin on the disc:

Since the disc is in equilibrium and the forces from the couple cancel out:

- Force = 50 N [1 mark]

[Total: 1 mark]

2a

(a)

(i) State the reason that the bus would topple to the left:

The bus begins to topple because:

- The line of action of the centre of gravity falls outside the base of the bus **OR** the anticlockwise moment (of the bus) is greater than the clockwise moment; [1 mark]

(ii) Explain what effect adding weights to the underside of the bus would be:

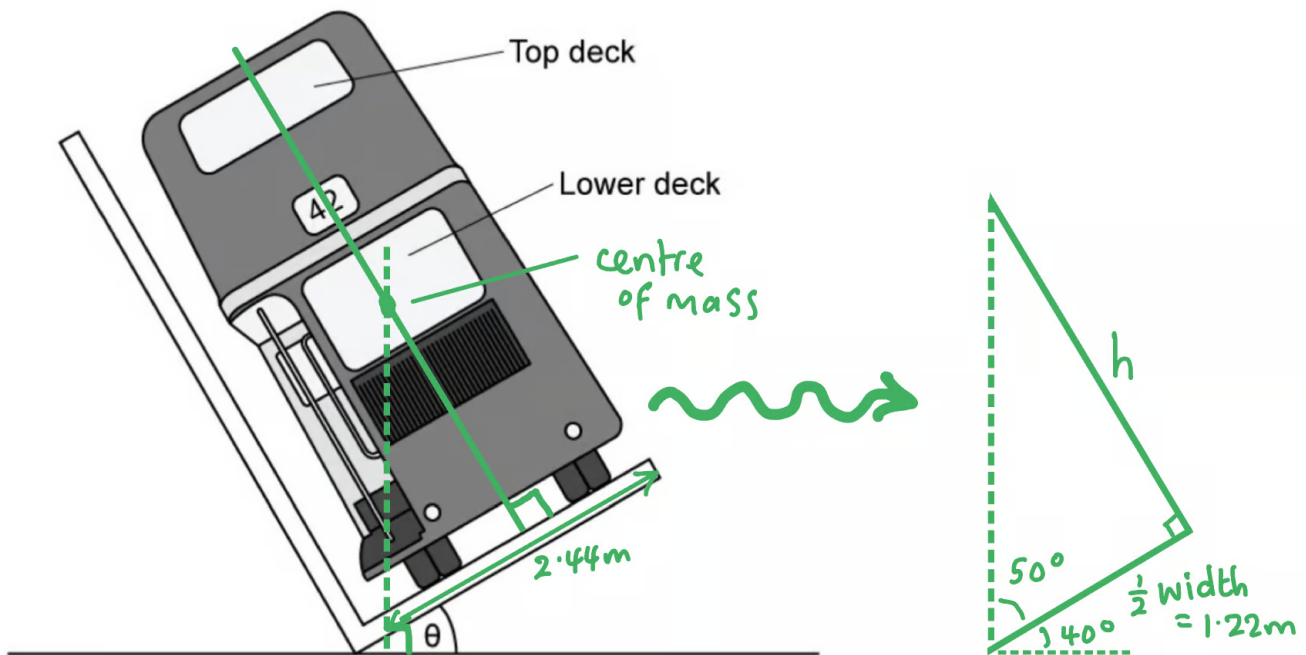
- Lower the centre of gravity; [1 mark]
- It could tilt more before toppling (to the left); [1 mark]
- Increases stability; [1 mark]

[Total: 4 marks]

2b

(b) Determine the height above the ground of the centre of mass of the bus when it is empty and normally parked:

Redraw the diagram as a triangle:



Either on the diagram or seen in later calculation

- Angle to be used in calculation = 50° **OR** short side of triangle = 1.22 m [1 mark]

Calculate the height of the centre of mass:

- $h = 1.22 \times \tan(50) = 1.5 \text{ m}$ [1 mark]

[Total: 2 marks]

Since A-level questions are more complex there are often 'intermediary' or 'working' marks for numbers seen within a calculation. In this case you are being awarded one mark for correctly deriving at least one of the values you will need to calculate.

Your working is being rewarded so it's well worth showing it.

The bus is not of uniform density, so its centre of mass will not be in the centre of the object.

2c

(c)

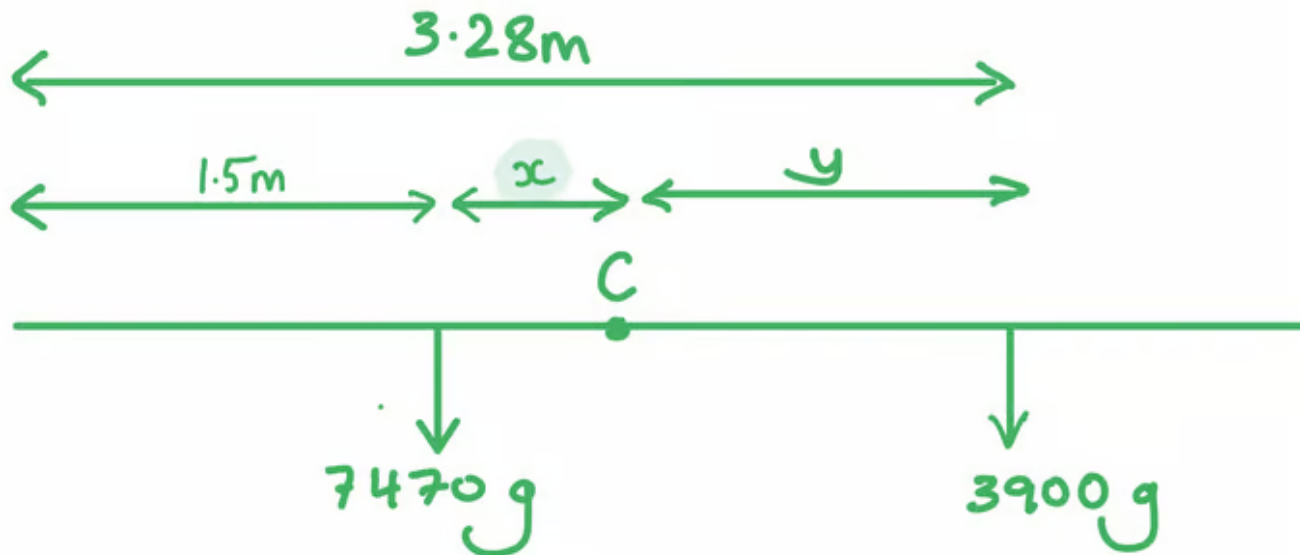
(i) Calculate the new height of the centre of mass of the bus:

Determine the total mass of the 'passengers' and the empty bus:

- Mass of 'passengers', $m_p = 65 \text{ kg} \times 60 \text{ passengers} = 3900 \text{ kg}$
- Mass of empty bus, $m_b = 7.47 \text{ tons} \times 1000 \text{ kg} = 7470 \text{ kg}$

Model the bus as a horizontal rod and take moments:

Method 1: take moments around the new centre of mass and calculate:



Set up simultaneous equations for x and y :

- In equilibrium, $7470g \times x = 3900g \times y \rightarrow 7470x = 3900y$ (equation 1)
- Taking distance measurements, $1.5 + x + y = 3.28 \rightarrow y = 1.78 - x$ (equation 2)

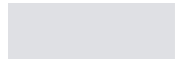
Substitute *equation 2* into *equation 1* and calculate:

- $7470x = 3900y$
- $7470x = 3900 \times (1.78 - x)$
- $7470x + 3900x = 6942$
- $x = 0.61 \text{ m}$ [1 mark]

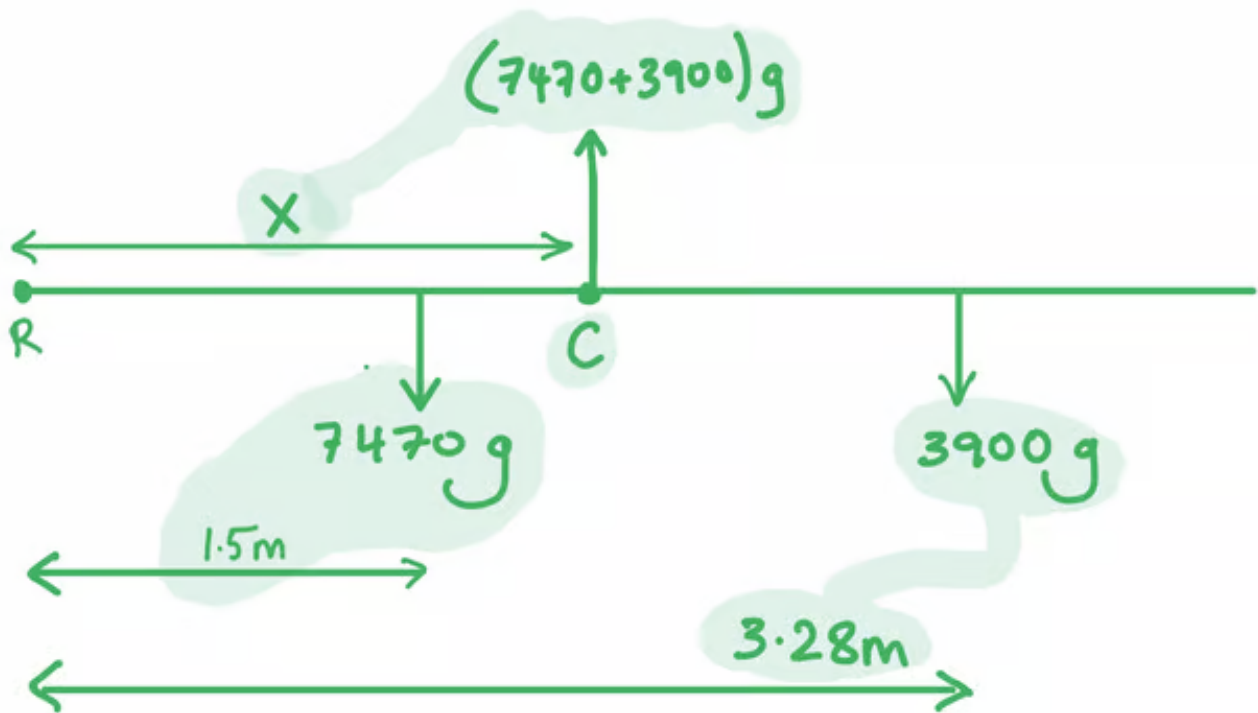
Add distances together to find the height of C above the road:

- new height, $h = 1.5 + 0.61 = 2.11 \text{ m}$ [1 mark]

Method 2: take moments about the road, assuming a combined mass, and



Method 2: take moments about the road, assuming a combined mass, and calculate:



Apply the principle of moments around R::

- In equilibrium, $(7470\text{ g} \times 1.5) + (3900\text{ g} \times 3.28) = (7470\text{ g} + 3900\text{ g}) \times X$ [1 mark]

Calculate:

- $(11\,205\cancel{\text{ g}}) + (12\,792\cancel{\text{ g}}) = (11\,370\cancel{\text{ g}}) \times X$
- $X = 2.11\text{ m}$ [1 mark]

(ii) Suggest why the test is done with the top deck weighted to capacity and the bottom deck left empty:

- This moves the centre of mass to the highest position / makes the bus the least stable it could be; [1 mark]

[Total: 3 marks]

2d

(d)

(i) Deduce whether the bus has passed the safety test:

Set up a triangle to calculate the angle using trigonometry:



- Angle inside triangle, $\theta = \tan^{-1}\left(\frac{2.11}{1.22}\right) = 60^\circ$
- Angle between bus and road, $\theta_R = 90 - 60 = 30^\circ$ [1 mark]

Draw a conclusion:

- The bus tilted to more than 28° without tipping so yes it does pass the safety test; [1 mark]

(ii) Explain whether all the passengers upstairs standing will make the bus fail its safety test.:

Calculate new angle between bus and road by moving the centre of mass 40 cm higher:

- $\theta = \tan^{-1}\left(\frac{2.11 + 0.4}{1.22}\right) = 64^\circ$

- New angle between bus and road, $\theta_R = 90 - 64 = 26^\circ$

Draw a conclusion:

- The bus tilts to less than 28° without tipping so yes it will fail the safety test; [1 mark]

For the final part of the question you could have recalculated the new centre of mass due to the passengers standing up. However, for only one mark, there is probably a quicker way.

By simply moving the centre of mass up by the full 40 cm (which is the maximum it could possibly shift) we still find that the bus fails the safety test, so the question has been answered without the need for further calculation.

3a

(a) Explain why the vessel rotates around its axis:

i) Refer to energy stores and transfers:

Any **three** from:

- Chemical energy store converted to thermal energy store as the fire burns; [1 mark]
- The water boils (undergoing) vaporisation / turns to steam; [1 mark]
- Nozzles create jets of steam which are perpendicular to the line between them; [1 mark]
- Thermal energy store transfers to kinetic energy; [1 mark]

ii) Refer to forces, naming any relevant laws:

Any **five** from:

- Steam exerts a force on the air therefore air exerts a force on the steam; [1 mark]
 - According to Newton's third law; [1 mark] (*only if linked to previous statement*)
- Creates unbalanced force which accelerates the vessel; [1 mark]
 - According to Newton's second law / $F = ma$; [1 mark] (*only if linked to previous statement*)
- Creates a couple; [1 mark]
- Turning force has torque; [1 mark]
- Torque = one of the forces $\times$ distance between the nozzles / force; [1 mark]

[Total: 8 marks]

3b

(b) Explain why simply puncturing holes in the can will not produce the rotation seen in the original Hero's engine:

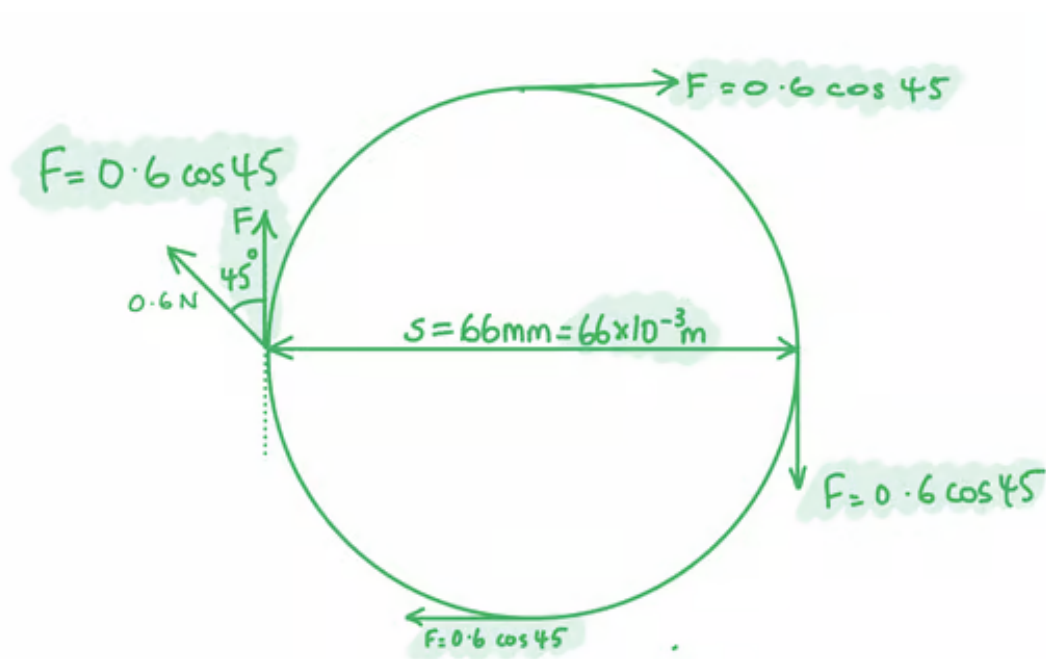
- The turning force is caused by a couple; [1 mark]
- Must be a pair of forces; [1 mark]
- Forces must be perpendicular to the distance between them; [1 mark]
- If the steam comes straight out / radially / not at an angle the component of the perpendicular force will be zero; [1 mark]

[Total: 4 marks]

3c

(c) Calculate the torque experienced by the can:

Make a simplified sketch of the situation:

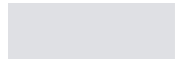


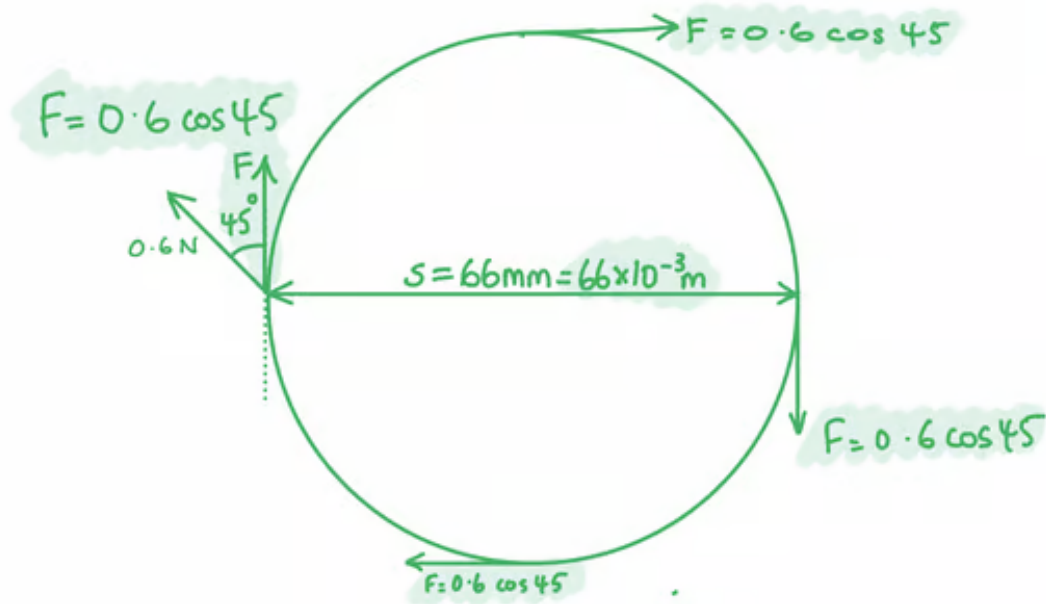
Calculate the component of the force which acts perpendicular to the line connecting the couples:

- $F = 0.6 \cos 45 = 0.424 \text{ N}$ [1 mark]

Calculate the torque produced by each couple:

- $\tau_c = F \times s = 0.424 \times (66 \times 10^{-3})$ [1 mark]





Calculate the component of the force which acts perpendicular to the line connecting the couples:

- $F = 0.6 \cos 45 = 0.424 \text{ N}$ [1 mark]

Calculate the torque produced by each couple:

- $\tau_c = F \times s = 0.424 \times (66 \times 10^{-3})$ [1 mark]

- $\tau_c = 2.80 \times 10^{-2} \text{ Nm}$ [1 mark]

Double the answer to get total torque, since there are two pairs of forces:

- $\tau_t = 2 \times (2.80 \times 10^{-2}) = 5.6 \times 10^{-2} \text{ Nm}$ [1 mark]

[Total: 4 marks]

3d

(d) Discuss an adjustment that could be made to the can to make it spin more quickly:

- Change the angle of the holes / jets; [1 mark]
- So the angle between the jet and can is smaller / less than 45° ; [1 mark]
- To increase the component of the force that acts in the perpendicular direction; [1 mark]

mark]

[Total: 3 marks]

